

CLASS 04 / SEP 15

GEOMETRY FROM SCRATCH

P1 CONTINUES — WHAT A SOLID IS MADE OF, UNDERNEATH

P1a DUE

P1b + P1c ISSUED TODAY

POINT → LINE → EDGE → FACE → SOLID

Five steps, and each one is a list of the thing before it.

POINT	LINE	EDGE	FACE	MESH
three numbers	two positions	two vertices	ordered corners	many faces
[x, y, z]	[p1, p2]	[v1, v2]	[v1, v2, v3]	[f1, f2, f3...]

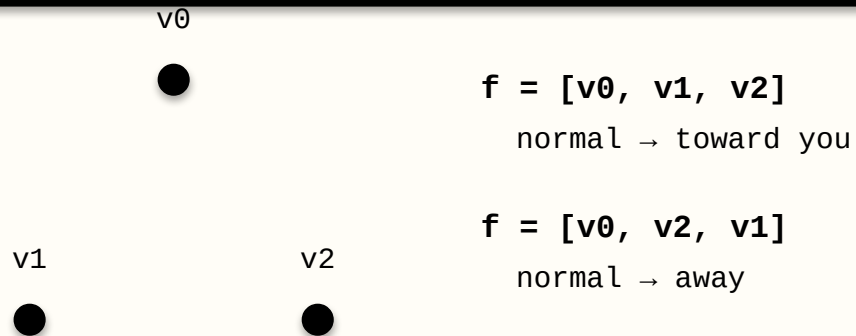
A POSITION IS NOT A DISPLACEMENT. THE SAME THREE NUMBERS MEAN DIFFERENT THINGS DEPENDING ON WHICH ONE YOU MEANT.

Nothing here is new geometry. It is the same collection, read at a different level: a face is a list of vertices, a mesh is a list of faces, and in Week 6 an array will be a list of copies.

A FACE IS AN ORDERED LIST OF CORNERS

Same three vertices, two different faces. The order is the difference.

WINDING ORDER



WHY IT MATTERS LATER

A flipped face tells a renderer the inside is the outside.

It tells a slicer there is nothing to fill.

Most failed prints this semester will start here, not at the printer.

CLOSED MEANS EVERY EDGE IS SHARED BY EXACTLY TWO FACES. CHECK IT BEFORE YOU BELIEVE IT.

THE PYRAMID

Five vertices, eight edges, five faces. Built by hand, driven by three numbers.

WHAT YOU ASSEMBLE

5

VERTICES

four base corners and an apex

8

EDGES

four around the base, four to the apex

5

FACES

four triangular sides and the base

EXPOSED INPUTS

Width

Depth

Height

Three numbers, and the whole solid rebuilds. That is the entire point of the exercise.

DO NOT FORGET THE BASE FACE. A PYRAMID WITH FOUR SIDES AND NO BOTTOM IS NOT A SOLID.

AND THEN IT GOES BACK INTO THE CSG FILE

The pyramid is not an exercise. It is the cutter for the study you already built.

01 BUILD IT

Five vertices, five faces, closed, with Width / Depth / Height exposed.

02 SUBTRACT IT

Cube minus pyramid. The same Boolean from Week 2, with your own solid.

03 DRIVE IT

Change the three inputs and show at least three settings of the result.

04 EXPLAIN IT

Face order, normals, closure — why it cuts cleanly instead of failing.

THIS IS WHY P1 IS ONE PROJECT AND NOT THREE ASSIGNMENTS.

Every project in this course does the same thing — the geometry you make in one milestone becomes the input to the next. You will never start from an unrelated object.

THE SAME THING, WRITTEN OUT

A mesh is two lists. That is the entire idea, and it fits on one slide.

```
verts = [[-w, -d, 0],          # 0
          [ w, -d, 0],          # 1
          [ w,  d, 0],          # 2
          [-w,  d, 0],          # 3
          [ 0,  0, h]]          # 4 apex

faces = [[0, 1, 4], [1, 2, 4],
         [2, 3, 4], [3, 0, 4],
         [0, 3, 2, 1]]          # the base – do not forget it

pyramid = mesh(verts, faces)
result  = difference(cube, pyramid)
```

WHAT IT MAPS ONTO

point	edge
vector	face
parameter	mesh
vertex	solid
line	normal

verts IS A LIST OF POSITIONS. faces IS A LIST OF INDICES INTO IT.

ISSUED TODAY

P1b — PANEL FROM SCRATCH

DUE WEEK 6 — Sep 29 / 7%

Find a panel in a building you like, keep it simple, and build it out of two lists - points and faces. It is the component you will array in P2 and the object you will print, so choose something that tiles.

FIND

A PANEL

One precedent, simple. It has to sit beside itself and close.

DECIDE

WHAT MOVES

The parameters, first. One has to change its shape, not its box.

BUILD

FROM TWO LISTS

Every point an expression of those numbers. Never a typed coordinate.

SHOW

THE RANGE

Low, middle, high - still closing, still tiling. P2b drives it.

REQUIREMENTS — P1b

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Find one panel in a built precedent and keep it simple. Choose it by one test: could you set it beside three copies of itself, edge to edge, and have the grid close? Name the building, the architect and your source on the sheet.

02

Decide the parameters before you model anything, and say in one line what each does. At least one must change the panel's shape — an aperture, a fold depth, a corner offset — not its bounding box.

03

Rebuild it from scratch: an explicit point list and an explicit face list, every point an expression of those parameters rather than a typed coordinate. Nothing collapsed, and it closes.

04

State each parameter's working range — where the panel still closes and still tiles — and show it at the low, middle and high end of the one an attractor will drive in P2b.

05

One 17 × 11 inch sheet, and one image on it: the panel rendered on a ground plane, one material, studio light, no context and no entourage. The object, its structure and nothing else.

06

Print the point list, the face list and the parameter table beside the render, and annotate the vertex order of one face. Explain normals, closure, and what breaks when a face is wound backwards.

ALSO ISSUED

P1c — FIRST PRINT

DUE WEEK 10 — Oct 27 / 7%

The same geometry, made real. A printable variant of the project you have been building since Week 2 — presented as a physical object at the midterm.

LIMIT

60 MM, 200 G

Each axis within 60 mm. The allocation includes failures.

RULE

NO SUPPORTS

Orientation is your only lever. Change the geometry if you must.

DUE

AT THE MIDTERM

The queue has latency the two-week cycle does not.

REQUIREMENTS — P1c

Fabrication milestone — visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements 10%.

01

Develop a printable variant of the geometry project, within 60 mm on each axis and the 200 g allocation, with no supports.

02

Present the physical print at midterm with one orientation/process sheet and a photograph.

03

Submit the editable source, exported mesh and slicer project. Explain any geometry changes made for printability.

CHECKPOINT · WEEK 6 — Sep 29 — Review print-ready geometry, physical units, orientation and the sliced preview before queueing.

NOW — BUILD THE PYRAMID

Rest of the session: from a single point to a closed solid that cuts.

✓

One point placed from an XYZ vector, and you can say why it is a position

✓

Four base corners and an apex, with Width / Depth / Height exposed

✓

All five faces joined — four sides and the base

✓

Closure checked: every edge shared by exactly two faces

✓

Cube minus pyramid working, off the same three inputs

✓

Your own panel named, and the numbers that will drive it written down

LEAVE WITH A SOLID THAT CUTS. THE SHEET IS THE EASY PART.

BEFORE NEXT CLASS

CONTINUE

U03 runs into next class. Build the panel from its own point and face lists; arrays start in Week 6, once the panel closes.

FINISH

P1a is due today. P1b — the panel sheet — is due Week 6, two weeks from now.

FIND

The panel. One building, one repeating component, simple enough to write as two lists, and it has to sit beside itself without a gap. Name the building and your source, and bring its point and face lists started - next class we build it.

CHECK

Your FabLab orientation now. P1c print-ready geometry is reviewed in Week 6.